in the figure is one hour behind the top clock. And this time lag is continuing to grow.

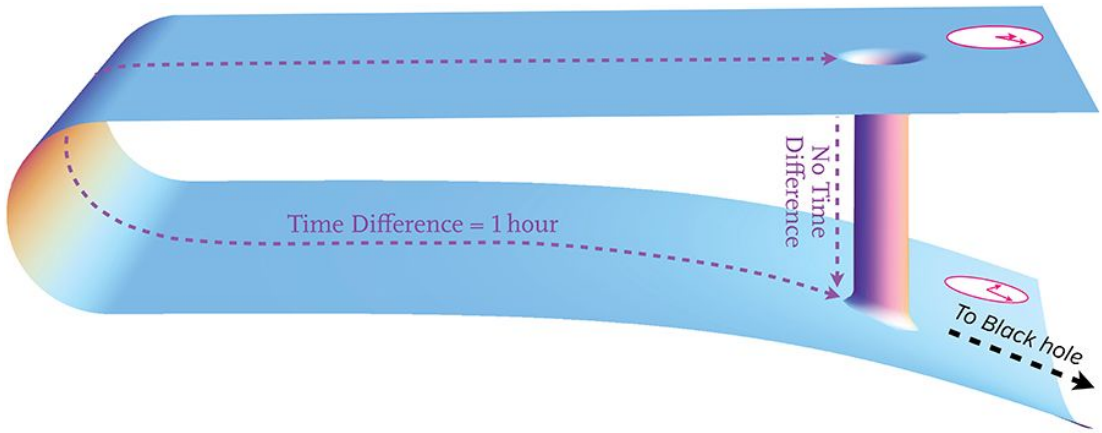


Fig. 30.3. Wormhole as a time machine.

Since there is only a tiny gravitational pull inside the wormhole, Einstein's law of time warps dictates that, as seen through the wormhole, time flows at essentially the same rate in the upper mouth as in the lower mouth. So there is no time lag when the clocks are compared through the wormhole. They are synchronized.

Suppose, further, for concreteness, that the distance from mouth to mouth in the external universe is short enough that you can traverse it in five minutes as measured by the clocks, and you can travel through the wormhole in one minute. Then this wormhole has already become a time machine. You leave the upper mouth at time 2:00 as measured by the clock there, and travel through the external universe to the lower mouth, arriving at 2:05 upper clock time and 1:05 lower clock time. You then make a one-minute trip upward through the wormhole, from lower mouth to upper. Since the clocks. are synchronized through the wormhole, you reach the upper mouth at time 1:06 as seen by both clocks. You arrive back at your starting point fifty-four minutes before your 2:00 departure, and you meet your younger self.